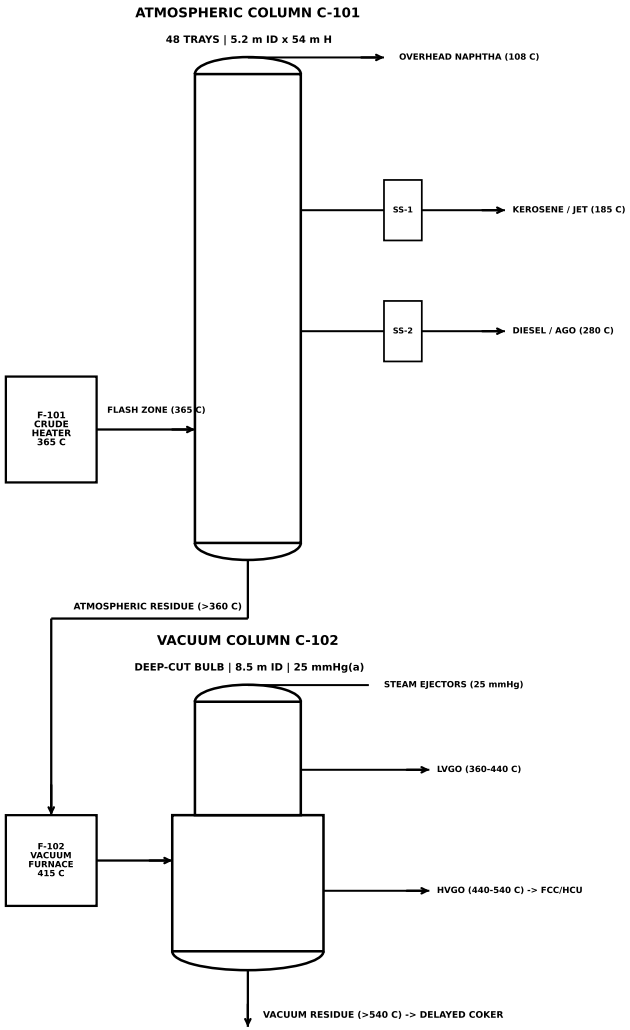
CRUDE FRACTIONATION & CUT-POINT TEMPERATURE PROFILES (CDU C-101 / VDU C-102)

TRUE BOILING POINT (TBP) CUT ARCHITECTURE | CRUDE HEATER F-101 & VACUUM EJECTOR CONDENSER SYSTEM



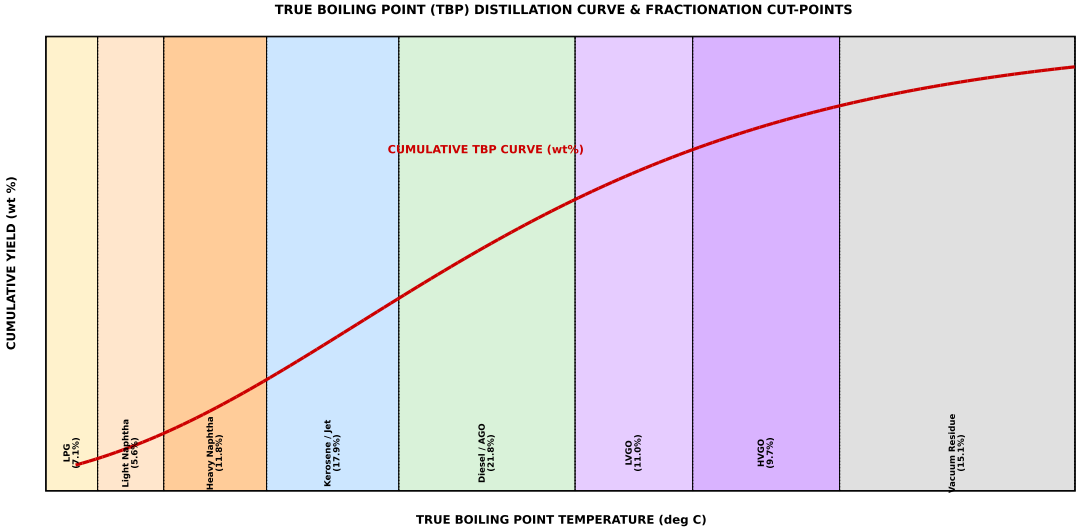


TABLE 1: PRIMARY FRACTIONATION CUT-POINTS & STREAM YIELDS

FRACTION STREAM	CUT T (C)	YIELD (wt%)	MASS (kg/h)	API	DOWNSTREAM UNIT
Off-Gas & LPG	< 35 C	7.06 %	39,858	95.0	Amine Treating -> LPG Spheres
Light Naphtha	35 - 80 C	5.56 %	31,417	72.0	Isomerization Unit (RON 88)
Heavy Naphtha	80 - 150 C	11.83 %	66,816	54.5	NHT -> Platformer (RON 102)
Kerosene / Jet A-1	150 - 240 C	17.93 %	101,220	42.0	Merox Sweetening -> Jet Pool
Diesel / AGO	240 - 360 C	21.78 %	123,012	34.0	Diesel HDS -> ULSD Blend
Light Vacuum Gas Oil	360 - 440 C	10.98 %	61,993	26.0	Hydrocracker / Diesel Blend
Heavy Vacuum Gas Oil	440 - 540 C	9.66 %	54,526	20.5	**FCCU (60%) & HCU (40%) Feed**
Vacuum Residue (VR)	> 540 C	15.20 %	85,828	5.5	**Delayed Coking Unit (DCU)**

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS PLANT DIGITAL TWIN SIMULATION

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
CDU & VDU FRACTIONATION & CUT-POINT PROFILES	PROCESS PLANT DIGITAL TWIN & CAD	B.E. CHEMICAL ENGINEERING	DIAGRAMMATIC / GRAPHICAL
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/VP/DWG-02	19 / 10 / 2026

VIRTUAL PETROLEUM REFINERY | 100,000 BPD CRUDE DIET | GAZEBO & ROS 2 DIGITAL TWIN | API / ASME BPVC